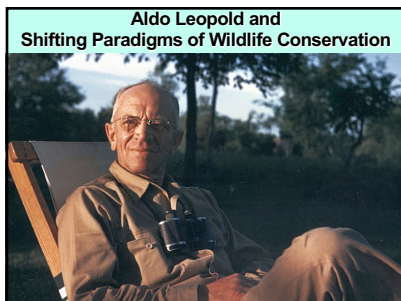
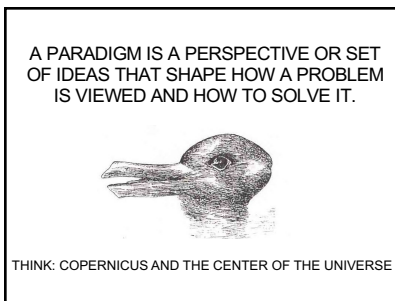
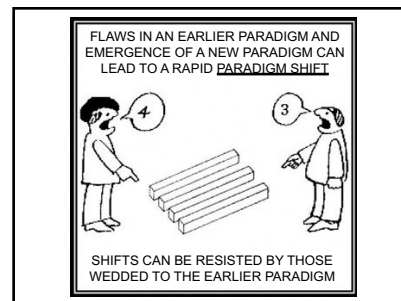




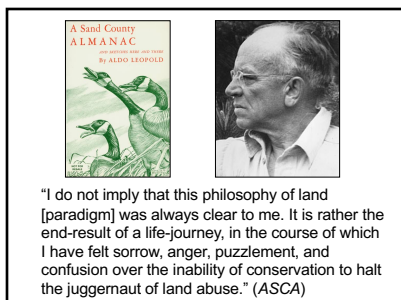
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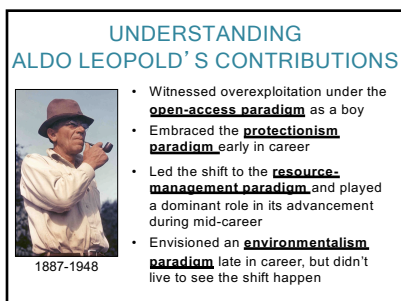
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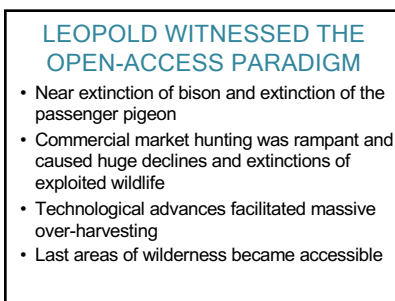
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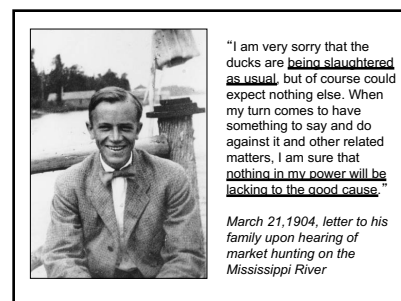
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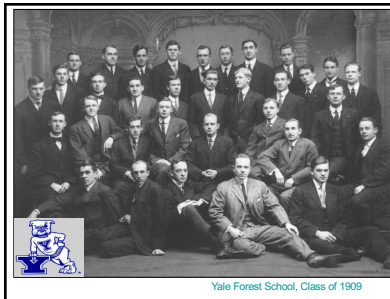


9

PARADIGM SHIFTS TO PROTECTIONISM

- Response to problems of open access was legal protection of wildlife resources (from direct taking) and suppression of factors harming valuable resources (e.g., fire and predators)
- 1900: Lacey Act tried to curb market hunting; National Audubon Society formed to oppose market hunting of birds
- 1903: First National Wildlife Refuge protected
- 1918: Migratory Bird Treaty Act (first federal wildlife protection!)
- State game protection programs expand rapidly

10



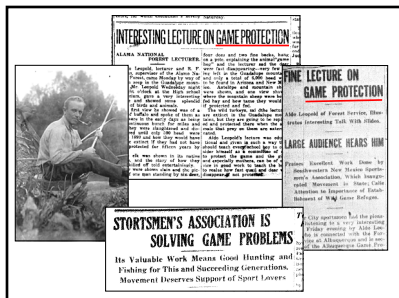
Yale Forest School, Class of 1909

11

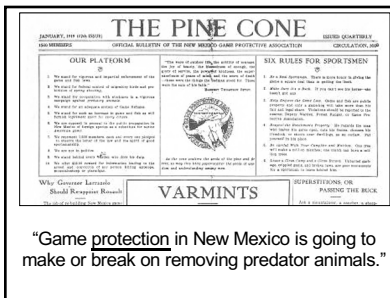
NATIONAL FORESTS IN THE SOUTHWEST 1909-1924 U.S. FOREST SERVICE



12

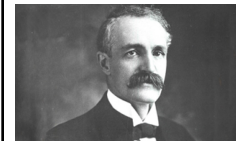


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14

GIFFORD PINCHOT AND MANAGEMENT OF FOREST RESOURCES

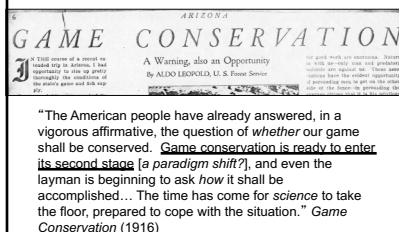


First Chief of the new US Forest Service

Coined the term "wise use" in 1910 to describe his concept of sustainable management [conservation] of natural resources on National Forests.

15

A SHIFT BEGINS TAKING SHAPE



16

ESSENCE OF A PARADIGM SHIFT: APPLYING SCIENCE-BASED MANAGEMENT TO GAME

"Game management problems are absolutely parallel to forest management problems, and the underlying principles are not merely parallel, they are identical." *Forestry and Game Conservation* (1918)

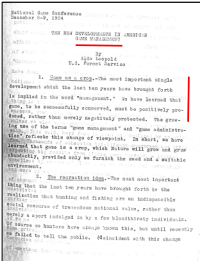
17

LEOPOLD RETURNS TO THE MIDWEST (1924)



US Forest Products Lab in Madison, Wisconsin

18

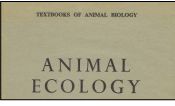


CONTRASTING OLD AND NEW PARADIGMS

"...game, to be successfully conserved, must be positively produced, rather than negatively protected." (1924)

19

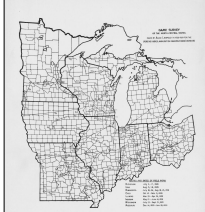
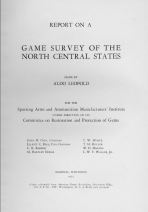
MAKING THE LINK BETWEEN ECOLOGY AND CONSERVATION

Charles Elton and Aldo Leopold

20

1928-31 GAME SURVEY OF THE NORTH CENTRAL STATES

21

LEOPOLD ADVOCATES FOR ACTIVE MANAGEMENT RATHER THAN PASSIVE PROTECTION

"The one and only thing we can do to raise a crop of game is to make the environment more favorable. ...It is the fundamental truth which the conservation movement must learn if it is to attain its objective." *Report of the Committee on American Wild Life Policy* (1929)

22

HERE'S THE PARADIGM SHIFT!

"Game conservation is at this moment in a particularly difficult stage of its development [a paradigm shift]. The set of ideas which served to string out the remnants of the virgin game supply [the protectionism paradigm], and to which many of us feel an intense personal loyalty, seem to have reached the limit of their effectiveness. Something new [the resource-management paradigm] must be done." *American Game Policy in a Nutshell* (1930)

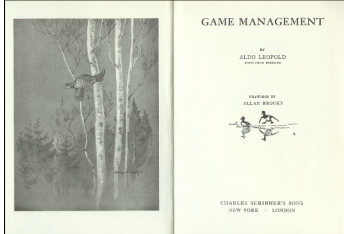
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THE SHIFT TO AN ECOLOGICALLY BASED, RESOURCE-MANAGEMENT PARADIGM

- Protection alone became recognized as inadequate
- Emphasis still on game species (resources)
- Shift toward ecologically based management
- 1930: "American Game Policy" published
- 1933: Aldo Leopold's *Game Management*
- Funding for management comes mainly from resource users

24

GAME MANAGEMENT (1933)



25

ECOLOGICAL PRINCIPLES INFORM MANAGEMENT

"The subject matter of this book has been hung upon a framework of 'factors,' rather than of species or land units, because the object is to portray the mechanism which produces all species on all lands, rather than to prescribe the procedures for producing particular species or managing particular lands." *Game Management* (1933)

26

LEOPOLD, THE UW PROFESSOR, THE FIRST IN THE NEW FIELD





RESEARCH OUTREACH INSTRUCTION
(1933-48)

27

FROM "GAME" TO "WILDLIFE"

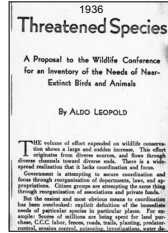
"Songbirds and other non-game wildlife are not yet thought of as subject to management. The fact that the objective is something to shoot, catch, or trap is, from the scientific point of view, irrelevant." (The University and Conservation of Wisconsin Wildlife 1937)



"My former two-credit course, 'Survey of Game Management', [will be replaced by] a new three-credit course to be called 'Wildlife Ecology'." (1938)

28

THREATENED AND ENDANGERED SPECIES



- "We have no right to exterminate any species. I stand on this as a fundamental principle."
- "...the immediate needs of our threatened fauna and flora must be addressed now before it is too late."

29

"LAND" AS A FOCUS FOR AN EMERGING PARADIGM SHIFT



"We have seen the emergence of two kinds of conservation and two systems of thought on the subject. One kind feels a primary interest in some one aspect of land (such as soil, forests, game or fish) with an incidental interest in the land as a whole. The other feels a primary interest in the land as a whole, with incidental interest in its component resources." *Conservation in Whole or in Part?* (1944)

30

The Pine Cone vs. A Sand County Almanac

- "The sportsmen demand the eradication of lions, wolves, coyotes, and bobcats. Either wolves or game must go." (1916)
- "Harmony with land is like harmony with a friend; you cannot cherish his right hand and chop off his left...you cannot love game and hate predators." (1949)

31

"LAND ETHIC," "ECOLOGICAL CONSCIENCE" AND "LAND HEALTH"



"A land ethic, then, reflects the existence of an ecological conscience, and this in turn reflects a conviction of individual responsibility for the health of the land." (ASCA)

32

HOW TO DECIDE WHAT IS RIGHT AND WRONG?



"...quit thinking about decent land-use solely as an economic problem. Examine each question in terms of what is ethically and aesthetically right, as well as what is economically expedient. A thing is right when it tends to preserve the integrity, stability, and beauty of the biotic community. It is wrong when it tends otherwise." (ASCA)

33

REALISTICALLY

- "Conservation is a state of harmony between men and land." (ASCA)
- "We shall never achieve harmony with the land, anymore than we shall achieve justice or liberty for all people. In these higher aspirations the important thing is not to achieve, but to strive..." (ASCA)



34

ALDO LEOPOLD'S CONTRIBUTIONS TO WILDLIFE CONSERVATION AS OF 1948

- Wildlife conservation became a science-based field built on ecological principles
- First generation of professional managers trained
- Better recognition of complex issues involved
- Encouraged connections with other conservation fields: forestry, agriculture, soil and water conservation, fisheries, land-use planning, etc.
- Introduced a more holistic vision for the future of wildlife conservation ("Conservation in Whole or in Part") which led to the environmentalism paradigm shift some two decades after his death

35

LEOPOLD'S LEGACY: ENVIRONMENTALISM

- Balancing relationship between humans and natural systems on which they depend in such a way that all the components remain intact and functioning [Sounds a lot like Leopold's "land health" concept!]
- Earth Day: April 22, 1970
- 1970s became "the environmental decade"
- Major environmental policies and laws reflect a new more holistic paradigm

36

THE PARADIGM SHIFTS!

- Resource-focused management alone recognized as insufficient to address complex problems facing the environment and non-resource wildlife
- Objectives of management expand beyond natural resources to the whole environment
- 1969: National Environmental Policy Act
- 1970: EPA formed
- 1970s: Non-game wildlife programs proliferate
- 1970s: Environmental policies and legislation expand
- 1973: Endangered Species Act
- By the 1980s: Biological diversity becomes a focus

37

CONSEQUENCES OF THE PARADIGM SHIFT FOR WILDLIFE CONSERVATION

- Maintaining biodiversity and ecosystem health became over-arching, unifying themes in wildlife conservation
- “The whole” is increasingly as important as “the parts” in policy and practice
- Leopold is widely recognized for his visionary ideas and contributions

38

END OF 20th CENTURY

- Wildlife conservation has broad public support
- Key laws and policies reflect public mandates
- Goals of wildlife conservation are widely understood and prioritized
- Guiding principles and enabling techniques are in place
- Scale of conservation activities is increasing
- Formal training and professionalism are the standard

39

PARADIGM SHIFT ON THE HORIZON?

- Erosion of the influence of wildlife science and expansion of influence of special interests on wildlife policies
- More controversies over wildlife conservation
- Changes in how wildlife conservation is funded
- Environmentalism hasn't halted ongoing loss of biodiversity or ecosystem deterioration
- Could this be leading to a paradigm shift?
- Could Aldo Leopold's land ethic be a central idea of a new paradigm?

40

“THROW YOUR WEIGHT AROUND”

- “I have no illusions about the speed or accuracy with which an ecological conscience can become functional...In such matters we should not worry too much about anything except the direction of travel. The direction is clear, and the first step is to *throw your weight around* on matters of right and wrong in land-use.
The Ecological Conscience (1947)

41